

Literature Review



Literature Review

CSE427: Machine Learning

Project Name: Student Stress Index Prediction

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1. Integrating MLOps with Machine Learning for Predictive Student Stress Analysis

Paper link: <https://ieeexplore.ieee.org/abstract/document/11210461>

Paper: [Drive link](#)

The paper is trying to predict the stress of the student from academic pressure and deadlines with the help of machine learning. This is relevant to our topic as we are also trying to solve the same problem. The dataset they are using contains 1100 student records with 21 integer features, including variables like anxiety, sleep quality, depression, and bullying. The authors implemented different types of machine learning algorithms including Logistic Regression, XGBoost, K-Nearest Neighbors, Random Forest, Voting Classifier and Gaussian Naive Bayes. Gaussian Naive Bayes achieved the highest predictive accuracy at 89.1%. Performance was evaluated using accuracy, precision, recall, and F1-score. Their results showed that ensemble-based models, particularly boosting techniques, exhibited greater predictive accuracy compared to traditional classification models. The integration of MLOps ensures the stress prediction model can process real-time data and continuously improve.

2. Predictive Modeling of Students' Stress Levels Using Machine Learning Algorithm

Paper link: <https://ieeexplore.ieee.org/abstract/document/10767502>

Paper: [Drive link](#)

The paper aims to predict and analyze the university student's stress levels using machine learning algorithms. This aligns with our goal of predicting student's stress index. The authors in this paper used a kaggle dataset which has 1100 student instances and 19 attributes covering academic, psychological, and environmental factors like student teacher relationship, anxiety and study load etc. The authors implemented several classification algorithms, including Random Forest, Decision Tree, K-Nearest Neighbors, and Naive Bayes, using a 70-30 training and testing data split. Performance was evaluated using accuracy, precision, and recall metrics. The Random Forest algorithm achieved the highest accuracy at 97.45%. One thing to notice is that the authors compared some of the previous works with theirs. Finally, the authors emphasized on using a diverse dataset with larger instances and features and also proposed the idea of parameter tuning to get better results.

3. The Role of Socioeconomic Factors in Machine Learning-Based Classification of Student Stress Levels

Paper link: <https://publications.essex.ac.uk/esj/article/id/453/>

Paper: [Drive link](#)

The study examined different ML models to determine if socioeconomic factors could be used to predict stress levels among students. The dataset that they are using contains 150 Bangladeshi undergraduate student survey responses with PSS-10 stress scale later cleaned and organized into csv files. Some of the features that they are using relate to our features such as family income, cgpa and scholarships. The authors tested 10 models including Naive Bayes, Random Forest, Decision Tree, CART, Logistic Regression, K-Nearest Neighbors, SVM, AdaBoost, and Bagging with the highest accuracy of 97.92% achieved by Decision Tree, CART, AdaBoost, and Bagging. They also used Accuracy, Precision, Recall and F1-score as performance metrics. The author then tested the models using only the socioeconomic factors but that didn't yield a very good result. This indicates we should not just focus on socioeconomic features to predict stress, but rather mix them with our academic features to get better results.

4. Academic Stress of Students in Higher Education using Machine Learning: A Systematic Literature Review

Paper Link: <https://ieeexplore.ieee.org/abstract/document/9974872>

Paper: [Drive link](#)

This paper is a systematic literature review of other research papers related to the stress prediction of students which can act as a foundation to understand the domain of stress prediction better. Instead of one dataset or papers, the authors analyzed 49 distinct research articles published between 2018 and 2023. The review found that Support Vector Machines, K-Nearest Neighbors, and Decision Trees are the most frequently used algorithms for stress prediction. Performance across the reviewed papers was mostly evaluated using classification and regression accuracy metrics. The paper shows that a student's program related information like their department is correlated with their stress level. The authors noted declining publications in developed economies and suggested optimizing models specifically for academic datasets, which are often small.

5. Predicting Depression in Bangladeshi Undergraduates using Machine Learning

Paper Link: <https://ieeexplore.ieee.org/abstract/document/8971369>

Paper: [Drive link](#)

This paper predicts depression among undergraduates from different universities. Stress and depression resembles connection since oftentimes stress leads to depression. The authors here collected data from 935 students using online surveys later cleaned and organized the raw data into csv files. The models used are Random Forest, Support Vector Machine, and K-Nearest Neighbors (K-NN). Performance was evaluated using accuracy, F-measure, precision, recall, and false negatives. Random Forest achieved the best results, yielding roughly 75% accuracy and a 60% F-measure. On the other hand, K-NN is not a good algorithm for prediction here due to a lower F-measure as K-NN works best with fewer features when classifying and the authors used 45 total features including the dummy variables. Random Forest produced lower false negatives. This is crucial because we want a lower false negative rate since we are working with a sensitive thing like stress and depression.

1. Data Analytics and Machine Learning Approaches for Predicting Academic Stress and Enhancing Student Wellbeing

Paper Link:

<https://fringe-global.com/ojs/index.php/ijari/article/view/machine-learning-for-predicting-academic-stress-and-student-well>

Paper: [Drive Link](#)

In this research, the researchers gathered a multimodal dataset containing information of 500 students across multiple institutions to build a machine learning model for predicting stress and enhancing student wellbeing. Their dataset included features like GPA, study hours, sleep duration, stress survey scores etc which explain student academic records, behavioural patterns and psychological matrices. To ensure privacy and integrity, the collected student data was securely stored on the blockchain network. They used machine learning classification models like Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost, and Neural Network (MLP). Among the preprocessed dataset of 500 students, they used 70% data for training, 15% for validation and 15% for final testing. After running all the algorithms, the XGBoost model gave the highest accuracy of 88.3%. They also evaluated the performance based on Precision, Recall, F1-score and ROC-AUC.

2. Predicting anxiety, depression, and insomnia among Bangladeshi university students using tree-based machine learning models

Paper Link: <https://onlinelibrary.wiley.com/doi/full/10.1002/hsr2.2037>

Paper: [Drive Link](#)

The research was conducted to predict mental health issues, specifically anxiety, depression and insomnia using three supervised, tree based machine learning classifiers: Decision Tree, Random Forest and XGBoost. The authors conducted a survey using google form distributed via social media and successfully gathered data from 1250 students across 24 different universities. They quantified the student's mental health using validated questionnaires like Generalized Anxiety Disorder-7 for anxiety, Patient Health Questionnaire-9 for depression and Insomnia Severity Index-7 for insomnia. They split the data taken into training and tests using 10-fold cross validation. In this method, they divided the data into 10 samples. In each iteration, nine samples were used as training data and one as testing data, and then the results were averaged. The data included Sociodemographic factors, social media addiction and morningness-eveningness. They evaluated the performance of the three models against the test data using accuracy, sensitivity, specificity, recall, precision, and Area Under the ROC Curve. Among the three models, XGBoost model gave the highest accuracy: 86% for anxiety, 84% for depression and 81% for insomnia.

3. Estimation of Student Stress Prediction Using Machine Learning for MCA Students Under Pune University

Paper Link:

<https://ijarcce.com/papers/estimation-of-student-stress-prediction-using-machine-learning-for-mca-students-under-pune-university/>

Paper: [Drive Link](#)

This research focuses on predicting stress level among the Master of Computer (MCA) students at Pune University. For that, the researchers collected data of 1000 students from their own university using google form and categorized the stress level into three groups — low, moderate and high. Their dataset includes demographic, academic, lifestyle and social factors such as assignment pressure, examination stress, sleep quality etc. They implemented four machine learning algorithms which are XGBoost, Random Forest, PCA-SVM and Logistic Regression. Among them XGBoost gave the highest accuracy which is 86%. Moreover, to measure the performance, they also calculated the Precision, Recall, F1-score and Balanced Accuracy.

4. Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students

Paper Link: <https://www.mdpi.com/2306-5354/10/5/575>

Paper: [Drive Link](#)

This research was conducted to predict mental health using various machine learning classifiers: Generalized linear models, K-Nearest Neighbour, Naive Bayes, Neural Networks, Random Forest, Recursive Partitioning, Bagging and Boosting. The researchers conducted an online cross-sectional survey and successfully gathered data from 15,366 university students across 17 different ASEAN University Network member universities. For the machine learning models to make predictions, the continuous scores from the K6 scale were converted into a simple binary variable: 1 (indicating a mental health problem) and 0 (indicating no problem). They partitioned the data 80% for training, and the rest 20% for testing. They also utilized 15 fold cross validation resampling which was iterated 10 times. The data included different demographic factors, physical activity, diet, tobacco use etc. They evaluated the performance of the models against the test data using accuracy, error rate, kappa, sensitivity, specificity, Area Under the ROC Curve (AUC), and Gini index. Among the models, the Random Forest classifier gave the highest performance: 90.1% for accuracy, 98.0% for sensitivity, and 0.966 for AUC. The accuracy they gained for such a huge amount of data is genuinely impressive and deserves praise, proving that this technology can be trusted to help students.

5. Comparison of the Performance of Machine Learning-based Algorithms for Predicting Depression and Anxiety among University Students in Bangladesh

Paper link:

https://journals.lww.com/shbh/fulltext/2022/05020/comparison_of_the_performance_of_machine.5.aspx

Paper: [Drive Link](#)

The research was conducted to predict mental illness, specifically depression and anxiety, using six machine learning classifiers: Logistic Regression, Random Forest, Support Vector Machine, Linear Discriminant Analysis, K-Nearest Neighbors, and Naïve Bayes. The authors conducted a structured questionnaire-based online survey and successfully gathered data from 2,121 students across private and public universities in Bangladesh. They quantified the students' mental health using validated questionnaires like the Patient Health Questionnaire-9 for depression and the Generalized Anxiety Disorder-7 for anxiety. They split the data taken into training and tests using a standard train-test split procedure. The data included sociodemographic variables and behavioral factors related to the first wave of the COVID-19 pandemic. They evaluated the performance of the six models against the test data using accuracy and various other performance parameters. Among the six models, the Random Forest model gave the highest accuracy for predicting depression (88.66%), while the Support Vector Machine model gave the highest accuracy for predicting anxiety (91.49%).

1. Mental Stress Detection in University Students using Machine Learning Algorithms

Paper Link:

<https://www.sciencedirect.com/science/article/pii/S1877050919306581>

Paper: [Drive Link](#)

In this paper, the dataset is a small collection of data of 206 students from a particular university. The main goal was to achieve whether the stress level is high or medium or low according to the 14 questions they were asked. Each question had 5 options. Based on that they will determine the chance of suicide. So, we are dealing with 206 rows and 14 columns of data with 5 variations in each column. This paper used Random Forest, SVM, Naive Bayes, KNN of machine learning to predict the outcome. The accuracy percentage of different methods is hereby, Random Forest – 83.33, Naive Bayes – 71.42, Support Vector Machine – 85.71, K-Nearest Neighbour – 55.55. In conclusion, Even with small datasets, but as it's from a single university, it is much likely to successfully predict the stress level of students within the university, Thus same approach can be made for different university to figure out their stress level.

2. Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring

Paper Link:

https://www.researchgate.net/publication/394292904_Protecting_Student_Mental_Health_with_a_Context-Aware_Machine_Learning_Framework_for_Stress_Monitoring

Paper: [Drive Link](#)

This paper aims to predict stress that do not show severely or before catastrophic medical conditions to prevent the inconvenience. This paper conducted their research based on 2 different datasets. One is 'student stress factor' dataset with 1100 data rows with 21 features. Another dataset is "Stress and well being data of college students" with 843 records and 26 features. Upon these datasets, six different classifiers was used. Such as, SVM, Random Forest, Gradient Boost-ing, XGBoost, AdaBoost, Bagging. To enhance performance, they used ensemble strategies, including hard voting, soft voting, weighted voting, and stacking. Among these models, the best model performance is 93.09% accuracy with weighted hard voting on 1st dataset. While the 2nd dataset provided 99.53 percent accuracy which is much higher than the benchmarks of the pasts. In conclusion, this paper introduced a context-aware machine learning with different models on different datasets with several data processing mechanism and performance enhancer to have a credible accuracy and legitimacy on predicting prior stress detecting index.

3. Classification of Student Stress Levels Using Machine Learning Methods

Paper Link:

https://www.researchgate.net/publication/396771472_Classification_of_Student_Stress_Levels_Using_Machine_Learning_Methods

Paper: [Drive Link](#)

The usual method to analyze stress index is to rely on surveys and subjective assessments based upon the responses from the students, but this paper considered the individual differences and cultural factors and manipulation of responses. This paper deals with the dataset from Kaggle consisting of 1100 data with 21 features, where the data is related to daily activities, age, gender, work, study, etc. The paper worked based on the target column named “stress level”. They split the dataset into 80 – 20, where 20 is for testing. The target column is scaled into 3 segments- low, medium, high. The dataset has a balanced class distribution, with no significant imbalance between. Necessary preprocessing was done on the dataset but handled the corruption of the dataset while preprocessing by using `df.isnull()`, `sum()` check etc. Data was analyzed with the help of confusion matrix. Three classifier were used for this paper, Such as, Logistic Regression with 88.1, Random Forest with 86.2 and XGBoost with 86.8 percent accuracy. In conclusion, this paper dealt with different features than traditional methods to predict a higher accuracy in finding stress index. Like thinking out of the box.

4. A Machine Learning based Approach for Mapping Personality Traits and Perceived Stress Scale of Undergraduate Students

Paper Link: <https://www.mecs-press.org/ijmecs/ijmecs-v11-n8/IJMECS-V11-N8-5.pdf>

Paper: [Drive Link](#)

This paper focused on personality traits based on a very small dataset of 150 participants. They surveyed 5 big personalities of individuals and tried to predict the stress level based on Sheldon Cohens' "Perceived Stress Scale". In this small dataset, nine different machine learning classifier models were applied. For performance evaluation, they have utilized precision, recall, f1-score, and accuracy. Among the classifiers, the best model is K-NN which provides 70% accuracy. The features or personality traits which are used are, Openness to experience, Extraversion, Agreeableness, Neuroticism And Conscientiousness. From the paper, it can be said that the personality index is not much of a good source for finding stress in students. Since most of the classifier models predicted poorly on figuring out the stress level. We can say it is better to use other features for finding the stress index of a student.

5. STRESS PREDICTION OF STUDENTS USING MACHINE LEARNING

Paper Link:

https://www.academia.edu/78914619/Stress_Prediction_of_Students_using_Machine_Learning

Paper: [Drive Link](#)

This paper used data from 200 students and used classification algorithms such as Naive Bayes , Linear Regression, Multi layer perceptron, Bayes Net, J48 and random forest to figure out the stress index of a student. The Authors used TP, FT, ROC, F-Measure and MCC, Precision for the performance parameters. Among the classifiers, Random Forest gives a high amount of accuracy of 94.73 percent. The paper considered different types of stress such as environmental, physiological and social stressor, Based on these, they tried to predict stress level to overcome many disease such as respiratory failure, diabetes, asthma and even suicide. The three categories in which the authors tried to class stress between are acute stress, episodic acute stress and chronic stress. This paper also used the PSS scale (The perceived stress scale) to assess the stress. The dataset is divided into 80-20 for the training and testing. The approach of this paper got higher accuracy in most of the classifiers where no classifier has an accuracy below 85 percent. In conclusion, even if the paper worked on only 200 students, the accuracy can justify the approach of the authors for a legitimate predictive model for other datasets.

Summarized Literature Review

Literature Review

There has been a lot of research on guessing how stressed students are using machine learning. Most of these studies look at a mix of school life, mental health, and the environment around the students. For example, research by S. Bhujbal et al. [1] looked at 1,100 student records. They wanted to see how academic pressure and tight deadlines affect students. They used Gaussian Naive Bayes to check 21 different things, like how well students sleep, if they face bullying, and their anxiety levels. This model gave them a good accuracy of 89.1%. Similarly, a study by K. Yeler and S. Sürücü [2] worked with a Kaggle dataset that also had 21 features. They looked at students' daily habits, age, and how they study. They divided the stress into three simple groups: low, medium, and high. After cleaning the data to remove any missing or broken values, they found that Logistic Regression was the best model, giving an 88.1% accuracy. Sometimes, researchers look at very specific groups of students. P. M. Ladkat and D. Gavhane [3] focused on Master of Computer Applications (MCA) students. These students face a huge amount of pressure from tough coding assignments and long exams. The researchers collected data from 1000 students using Google Forms. They asked questions about sleep quality and exam fear. They found that an XGBoost model was the best choice, achieving an 86% accuracy, mostly because XGBoost is really good at finding hidden patterns in survey answers.

Academic stress is often linked to bigger mental health problems like depression, anxiety, and lack of sleep. A study by A. A. Choudhury et al. [4] focused on 935 university students in Bangladesh. They collected data using online surveys and checked 45 different features, including living conditions and study pressure. They found that Random Forest was the best model, giving about 75% accuracy. They really liked this model because it had a very low false-negative rate. In simple words, a low false-negative rate means the machine rarely makes the mistake of saying a depressed student is completely fine. This is extremely important when dealing with sensitive issues like mental health, where missing a sick student can be very dangerous. They also noted that simpler models like K-Nearest Neighbors (KNN) do not work well when there are too many features to look at. Looking at a much bigger picture, M. Rosenberg et al. [5] did a massive survey with 15,366 students from 17 different universities in the ASEAN region. They checked personal habits like physical activity, diet, and tobacco use. To make sure their results were strong, they tested the data many times using a 15-fold cross-validation method. This means they trained and tested the machine over and over again on different parts of the data. Their Random Forest model achieved a very impressive 90.1% accuracy, which proves that machine learning can handle huge amounts of student data reliably and help universities support their students better.

Some researchers wanted to see if they could predict stress using only specific types of information instead of just grades. A study by A. A. Marouf et al. [6] tried to link the "Big Five" personality traits—like how outgoing, nervous, or open to new experiences someone is—to stress levels. They used Sheldon Cohen's well-known stress scale on 150 undergraduate students. However, their best model, K-Nearest Neighbors (KNN), only got a 70% accuracy. This shows that just knowing someone's personality is not enough to guess their stress accurately, because stress comes from outside pressures too. Similarly, T. Biswas et al. [7]

looked at 150 Bangladeshi students to see if socioeconomic factors, like family income and scholarships, could predict stress. When they used only these financial factors, the results were not very good. But when they combined the family background data with academic features like a student's CGPA, the models did much better. By using Decision Trees and AdaBoost, their accuracy jumped up to 97.92%. This makes perfect sense because financial problems can cause stress, which then ruins grades, creating a bad loop. Finally, to make predictions even more accurate, some researchers combine multiple models together. This is called an ensemble strategy. Instead of relying on one algorithm, it acts like a panel of expert judges voting on the final answer. By using methods like weighted voting or stacking different models, researchers have been able to push the accuracy of stress prediction incredibly high. This proves that combining different features and different algorithms gives the best picture of a student's mental well-being.

References:

- [1] S. Bhujbal, P. Lashkare, V. Dingalwar, A. Shende, R. Pawar and S. Kulkarni, "Integrating MLOps with Machine Learning for Predictive Student Stress Analysis," 2025 International Conference on Information, Implementation, and Innovation in Technology (I2ITCON), Pune, India, 2025, pp. 1-7, Available: <https://ieeexplore.ieee.org/abstract/document/11210461>
- [2] K. Yeler and S. Sürücü, "Classification of Student Stress Levels Using Machine Learning Methods," International Journal of Advanced Natural Sciences and Engineering Researches, vol. 9, no. 10, pp. 210-214, 2025. [Online]. Available: https://www.researchgate.net/publication/396771472_Classification_of_Student_Stress_Levels_Using_Machine_Learning_Methods
- [3] P. M. Ladkat and D. Gavhane, "Estimation of Student Stress Prediction Using Machine Learning for MCA Students Under Pune University," International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), vol. 14, no. 10, Oct. 2025. [Online]. Available: <https://ijarcce.com/papers/estimation-of-student-stress-prediction-using-machine-learning-for-mca-students-under-pune-university/>
- [4] A. A. Choudhury, M. R. H. Khan, N. Z. Nahim, S. R. Tulon, S. Islam and A. Chakrabarty, "Predicting Depression in Bangladeshi Undergraduates using Machine Learning," 2019 IEEE Region 10 Symposium (TENSYP), Kolkata, India, 2019, pp. 789-794, Available: <https://ieeexplore.ieee.org/abstract/document/8971369>
- [5] M. Rosenberg et al., "Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students," Bioengineering, vol. 10, no. 5, p. 575, MDPI, 2023. [Online]. Available: <https://www.mdpi.com/2306-5354/10/5/575>
- [6] A. A. Marouf, A. F. Ashrafi, T. Ahmed, and T. Emon, "A Machine Learning based Approach for Mapping Personality Traits and Perceived Stress Scale of Undergraduate Students," International Journal of Modern Education and Computer Science, vol. 11, no. 8, Aug. 2019. [Online]. Available: <https://www.mecs-press.org/ijmecs/ijmecs-v11-n8/IJMECS-V11-N8-5.pdf>

[7] T. Biswas, R. K. Mim, and F. S. Ferdous, "Machine Learning-Based Detection of Student Stress Levels Through Socioeconomic Status," Essex Student Journal, vol. 17, no. 1, Feb. 2026. [Online]. Available: <https://publications.essex.ac.uk/esj/article/id/453/>